Prediction Error Method for Identification of Subnetworks of Dynamical Systems ⋆

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Abstract: This paper considers prediction-error identification of a subnetwork embedded in a dynamic network. Measured boundary signals entering the subnetwork are appended to the exogenous inputs. Consistency requires that the part of these boundary signals generated by the subnetwork be reconstructible from the measured history; the subnetwork may be enlarged to obtain a simple structural sufficient condition. Under the stated model, stability, delay, disturbance-separation, informativity, and identifiability assumptions, the resulting criterion yields consistent estimates of the subnetwork transfer functions. The predictor may depend nonlinearly on the open-loop transfer functions, allowing consistent estimation with fewer measured internal signals than methods based on predictors that are linear in those transfer functions.

Keywords: System Identification, Networks, Dynamical Systems, Prediction error methods, Consistency, Informativity, Generic network identifiability.

1. INTRODUCTION

System identification for networks of dynamical systems is an active research field with a long history. Some early references are (Van Den Hof and Schrama, 1993; Van den Hof et al., 2013; Hendrickx et al., 2019). Much of the focus has been on identifying parts of a network, called a *subnetwork*, with the motivation that in practice it is impossible to identify large networks, and often only parts of a network are of interest in applications. However, it is important to recognize that the overall network will have an impact on how the subnetwork can be identified and what the quality of the subnetwork model will be. This impact is due to the feedback that the overall network usually provides to the subnetwork. This means that the problem of identifying subnetworks is a problem of system identification under feedback, a topic that also has a long history, see e.g. (Gustavsson et al., 1977). A more recent survey is (Forssell and Ljung, 1999).

We develop a prediction error method (PEM) for this problem. The main idea is to treat measured boundary signals entering the target subnetwork as additional measured inputs and then apply complete-network PEM.

Identification of individual modules and subnetworks in linear dynamic networks has received considerable attention. The local methods of (Van den Hof et al., 2013) construct predictors around a designated module, and (Dankers et al., 2016) give conditions for selecting subsets of node signals as predictor inputs. These methods target one designated transfer function, and their local predictor formulations are linear in that target open-loop module. Network abstraction provides another way to remove unmeasured node signals while preserving a designated target module (Weerts et al., 2020). Complete-network methods instead estimate all modules jointly (Weerts et al., 2018). More generally, (Hansson et al., 2025) establish consistency for complete dynamic networks with predictors that may depend nonlinearly on several open-loop transfer functions. This result provides the complete-network basis for the subnetwork method developed here.

In the standard closed-loop setting considered by (Ljung, 1999; Forssell and Ljung, 1999), the output entering the feedback loop is also the output being predicted. Their prediction-error analysis assumes that the plant contribution to the predicted output is independent of the current and future innovations. A delay in the loop is a common sufficient condition: the current prediction error cannot pass through the feedback loop and affect the input used to predict the same output.

This property is not automatic when only part of a dynamic network is measured. A boundary signal returning to the subnetwork may depend on an unmeasured subnetwork output and can therefore contain information that is absent from the measured output history. Existing local-module methods address this issue by enlarging the local predictor input and output sets so that the relevant

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feedback paths are accounted for (Van den Hof et al., 2013; Dankers et al., 2016; Weerts et al., 2020).

Our method uses a different enlargement rule. Additional modules from the original network are included in the subnetwork to be estimated until the part of each returning boundary signal generated by the subnetwork is measured or reconstructible from the measured history. All modules in the enlarged subnetwork are then estimated jointly, and the prediction-error orthogonality argument can be applied. Our work builds on (Hansson et al., 2025), which considers complete dynamic networks. If only one transfer function is of interest, the initial subnetwork may consist of its module, so single-module identification is included as a special case.

Because the complete-network predictor can depend nonlinearly on several unknown open-loop transfer functions, the interactions among modules in the enlarged subnetwork can be represented directly. This can reduce the number of internal signals that must be measured compared with local methods based on predictors that are linear in the target modules.

2. MODEL OF NETWORK

We consider linear models for the vector-valued input signals $u_k \in \mathbb{R}^n$, $r_k \in \mathbb{R}^{m_r}$ and $e_k \in \mathbb{R}^{m_e}$, and the vector-valued output signal $y_k \in \mathbb{R}^n$ defined by

$$y_k = \mathbf{G}(q)u_k + \mathbf{\Omega}_y(q)r_k + \mathbf{H}_y(q)e_k \tag{1}$$

where $\mathbf{G}$, $\mathbf{\Omega}_y$ and $\mathbf{H}_y$ are proper transfer function matrices in the forward shift operator, q. We assume that $\mathbf{G}(q)$ is strictly proper. The sequence e_k is temporally independent, has zero mean, and has constant diagonal positive definite covariance matrix Σ_e. Let $\mathbf{\Upsilon}$, $\mathbf{\Omega}_u$ and $\mathbf{H}_u$, of dimensions $n \times n$, $n \times m_r$ and $n \times m_e$, respectively, also be proper transfer function matrices in the forward shift operator. We define a feedback connection

$$u_k = \mathbf{\Upsilon}(q)y_k + \mathbf{\Omega}_u(q)r_k + \mathbf{H}_u(q)e_k. \tag{2}$$

These two equations define what is often called a *network* of dynamical systems. We assume that the closed-loop system is well-posed and stable.

Another popular network formulation considered in the literature is given by

$$w_k = \bar{\mathbf{G}}(q)w_k + \bar{\mathbf{\Omega}}(q)r_k + \bar{\mathbf{H}}(q)e_k \tag{3}$$

with $(\bar{\mathbf{G}}, \bar{\mathbf{\Omega}}, \bar{\mathbf{H}})$ also proper transfer function matrices in the forward shift operator. We assume that $\bar{\mathbf{H}}(q)$ is diagonal. It is often assumed that $\bar{\mathbf{G}}(q)$ has a zero diagonal. This can be cast in the form of (1) by taking $u_k = y_k = w_k$ and $\mathbf{G}(q) = \bar{\mathbf{G}}(q)$. A network on the form (1) and (2) can be written

$$\begin{bmatrix} y_k \\ u_k \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{G}(q) \\ \mathbf{\Upsilon}(q) & 0 \end{bmatrix} \begin{bmatrix} y_k \\ u_k \end{bmatrix} + \begin{bmatrix} \mathbf{\Omega}_y(q) \\ \mathbf{\Omega}_u(q) \end{bmatrix} r_k + \begin{bmatrix} \mathbf{H}_y(q) \\ \mathbf{H}_u(q) \end{bmatrix} e_k \tag{4}$$

which with $w_k = (y_k, u_k)$ and obvious definitions of transfer function matrices is of the form in (3) with a zero diagonal $\bar{\mathbf{G}}(q)$. Hence the two formulations can be considered to be equivalent, and we will use them interchangeably.

We assume that all the transfer functions we have defined are dependent on some parameter θ, parts of which are going to be identified. We denote with Θ_o the set of parameters θ such that the closed-loop system has all poles strictly inside the unit circle. We define the compact set $\Theta \subset \Theta_o$ such that it contains the parameter θ_0 which represents the closed-loop system from which we will collect data. This data will be defined later on. We assume that all transfer functions are rational functions in θ and q. If we want to emphasize the parameter dependence we may e.g. write $\mathbf{G}(q;\theta)$ for transfer functions and $y_k(\theta)$ for signals.

3. PREDICTION ERROR METHOD AND CONSISTENCY

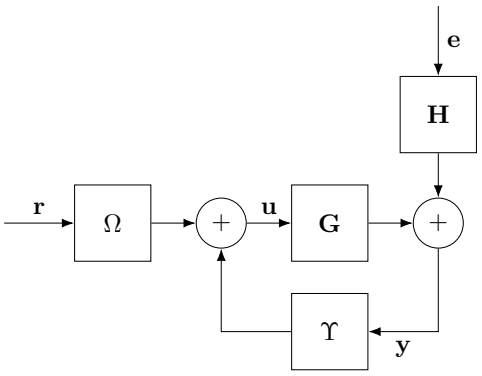


Fig. 1. Block diagram for a dynamic network.

It is possible to consider a problem equivalent to (1–2) which is of the same form but of higher dimension for which $\mathbf{G}$ is diagonal, Υ and $\mathbf{\Omega}_u$ are zero-one matrices and $\mathbf{\Omega}_y$ and $\mathbf{H}_u$ are zero. This is shown in Section 6. For that case we have for diagonal $\mathbf{H} = \mathbf{H}_y$ a multidimensional Box-Jenkins model in a feedback configuration as in Figure 1. In (Hansson et al., 2025) it has been shown how to obtain the predictor and formulate a maximum likelihood (ML) problem that can be solved reliably for multiple single-input single-output ARMAX models in this feedback configuration also when not all signals in the network are measured.

To estimate part of a network, partition it as

$$\begin{bmatrix} w_k^A \\ w_k^B \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{G}}_A & \bar{\mathbf{G}}_{AB} \\ \bar{\mathbf{G}}_{BA} & \bar{\mathbf{G}}_B \end{bmatrix} \begin{bmatrix} w_k^A \\ w_k^B \end{bmatrix} + \begin{bmatrix} \bar{\mathbf{\Omega}}_A \\ \bar{\mathbf{\Omega}}_B \end{bmatrix} r_k + \begin{bmatrix} \bar{\mathbf{H}}_A \\ \bar{\mathbf{H}}_B \end{bmatrix} e_k.$$

The signals and modules with superscript A form the A-part, and those with superscript B form the B-part.

We define $\hat{r}_k^A = (r_k, w_k^B)$, $\hat{r}_k^B = (r_k, w_k^A)$ and

$$\hat{\mathbf{\Omega}}_A = \begin{bmatrix} \bar{\mathbf{\Omega}}_A & \bar{\mathbf{G}}_{AB} \end{bmatrix}, \quad \hat{\mathbf{\Omega}}_B = \begin{bmatrix} \bar{\mathbf{\Omega}}_B & \bar{\mathbf{G}}_{BA} \end{bmatrix}.$$

Then it holds that

$$w_k^A = \bar{\mathbf{G}}_A w_k^A + \hat{\mathbf{\Omega}}_A \hat{r}_k^A + \bar{\mathbf{H}}_A e_k,$$
$$w_k^B = \bar{\mathbf{G}}_B w_k^B + \hat{\mathbf{\Omega}}_B \hat{r}_k^B + \bar{\mathbf{H}}_B e_k.$$

We then let $w_k^A = (y_k^A, u_k^A)$ and similarly for w_k^B and we also let

$$\bar{\mathbf{G}}_A = \begin{bmatrix} 0 & \mathbf{G}_A \\ \Upsilon_A & 0 \end{bmatrix}$$

and similarly for $\bar{\mathbf{G}}_B$. We then have two systems of the form in (4), each of which is equivalent to the form in (1–2). We may then use the result in Section 6 to deduce that they can be written as

$$\tilde{y}_k^A = \tilde{\mathbf{G}}_A \tilde{u}_k^A + \tilde{\mathbf{H}}_A e_k$$
$$\tilde{u}_k^A = \tilde{\Upsilon}_A \tilde{y}_k^A + \tilde{\Omega}_A \hat{r}_k$$
$$\tilde{y}_k^B = \tilde{\mathbf{G}}_B \tilde{u}_k^B + \tilde{\mathbf{H}}_B e_k$$
$$\tilde{u}_k^B = \tilde{\Upsilon}_B \tilde{y}_k^B + \tilde{\Omega}_B \hat{r}_k$$

for diagonal transfer function matrices and where the other matrices are zero-one matrices. Moreover, it holds for zero-one matrices (Φ_A, Φ_B) that

$$w_k^A = \Phi_A \tilde{y}_k^A, \quad w_k^B = \Phi_B \tilde{y}_k^B.$$

It then holds with $\tilde{\Omega}_A = \begin{bmatrix} \tilde{\Omega}_{AA} & \tilde{\Omega}_{AB} \end{bmatrix}$ and $\tilde{\Omega}_B = \begin{bmatrix} \tilde{\Omega}_{BB} & \tilde{\Omega}_{BA} \end{bmatrix}$ that

$$\tilde{\Omega}_A \hat{r}_k^A = \tilde{\Omega}_{AA} r_k + \tilde{\Omega}_{AB} \Phi_B \tilde{y}_k^B$$
$$\tilde{\Omega}_B \hat{r}_k^B = \tilde{\Omega}_{BB} r_k + \tilde{\Omega}_{BA} \Phi_A \tilde{y}_k^A.$$

If we define $\tilde{\Upsilon}_{AB} = \tilde{\Omega}_{AB} \Phi_B$ and $\tilde{\Upsilon}_{BA} = \tilde{\Omega}_{BA} \Phi_A$ we obtain the network in Figure 2, where we do not write out the tildes. The conclusion of this exercise is that there is no loss in generality to consider a feedback connection as in Figure 2 where $\mathbf{G}_A$ and $\mathbf{G}_B$ are diagonal and where the Υ and Ω matrices are zero-one matrices. Let x_k denote the measured part of $(\mathbf{y}_k^A, \mathbf{u}_k^A)$. All modules included in the A-part are estimated. The signals entering the A-part from the B-part are treated as additional measured input signals. The resulting equations have the form considered in (Hansson et al., 2025). More specifically, the top part of

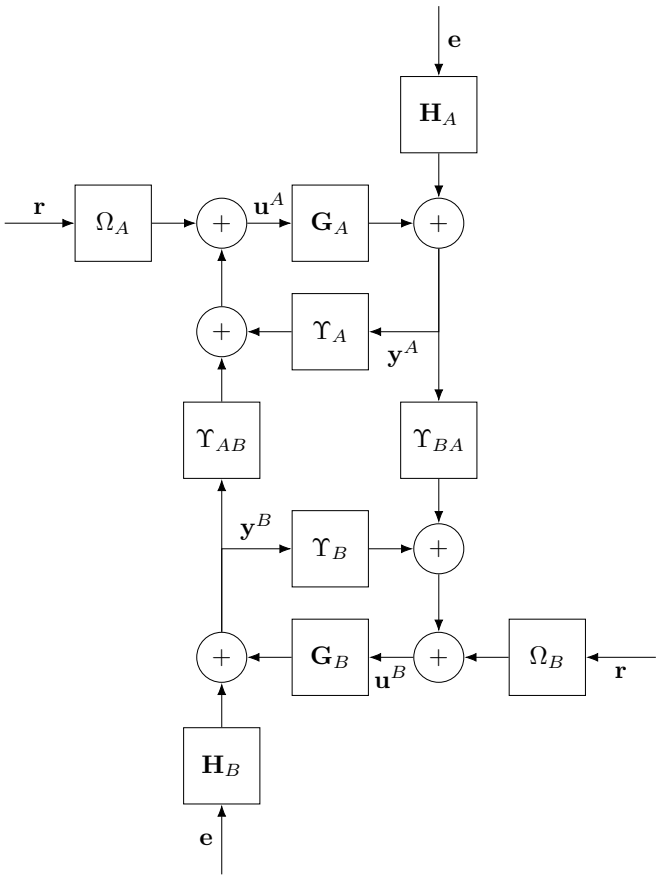


Fig. 2. Block diagram for a dynamic network partitioned into two subnetworks.

the network in Figure 2 can be cast in the form of the network in Figure 1 by replacing Ω with $\bar{\Omega} = [\Omega_A \ I]$ and r_k with $\bar{r}_k = (r_k, \bar{y}_k^B)$, where $\bar{y}_k^B = \Upsilon_{AB} y_k^B$ is a signal we assume we can measure.[1] This is visualized in Figure 3. It should be stressed that $\bar{y}_k^B$ will in general be

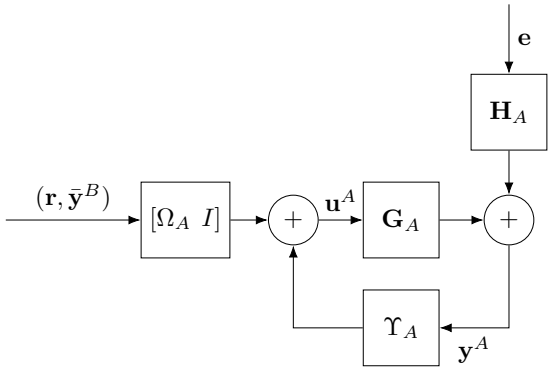


Fig. 3. Block diagram for a dynamic subnetwork.

correlated with e_k, and therefore the criterion function obtained by applying the approach of (Hansson et al., 2025) to a subnetwork is not, in general, a true ML criterion. We impose the following boundary-information condition: the component of $\bar{y}_k^B$ generated by signals from A is causally reconstructible from $\{x_j : j < k\}$ and the measured exogenous inputs available at time k. Thus, $\bar{y}_k^B$ provides no additional information about the current A-part innovation.

It is not necessary to measure every signal from A entering B. Only a boundary component that can affect $\bar{y}_k^B$ must be measured or causally reconstructible. Measuring every signal from A entering B is a simple, but potentially conservative, structural condition that ensures the boundary-information condition. If this structural condition does not hold, relevant modules can be moved from B to A and the condition checked at the new boundary.

Enlargement only moves modules from B to A and terminates after at most all modules in B have been included. In the worst case, estimation covers the complete network. Enlargement is needed only to obtain the simple structural condition; direct verification of reconstructibility may permit a smaller A-part. The method is most useful when a small measured or reconstructible boundary separates the target A-part from the B-part; otherwise, the number of nuisance modules and the optimization dimension can become large.

(Van den Hof et al., 2013; Dankers et al., 2016; Weerts et al., 2020) and the present method may enlarge the identification model beyond the original target. The former methods enlarge or abstract the predictor model around a designated module, whereas the present method enlarges the A-part by including original network modules until the boundary-information condition is satisfied. These enlargements need not contain the same transfer functions or require the same measured signals. The present method also allows the initial A-part to contain several target modules. If only one transfer function is the target, the initial A-part may contain only that module, and the method applies as a special case. Unlike the cited local

[1] We can equivalently assume that Υ_{AB} is known and that we can measure y_k^B.

predictor formulations, the resulting predictor may depend nonlinearly on the open-loop transfer functions and can consequently require fewer internal measurements. In the remaining part of this section we restrict ourselves to the case when $(\mathbf{G}_A, \mathbf{H}_A)$ defines ARMAX models. Notice that they are single-input single-output models, since the transfer functions are assumed to be diagonal.

To estimate the parameter vector θ for subnetwork A we consider the following optimization problem

$$\begin{aligned} \text{minimize} \quad & \frac{1}{2}\sum_{k=1}^{N}\left(\epsilon_k^T \Sigma_\epsilon^{-1}\epsilon_k + \ln\det\Sigma_\epsilon\right) \\ \text{subject to} \quad & \hat{x}_k = \mathbf{W}_x x_k + \mathbf{W}_r \bar{r}_k \\ & \epsilon_k = x_k - \hat{x}_k, \quad \theta \in \Theta \end{aligned} \tag{5}$$

with variables $(\theta, \epsilon, \hat{x})$, where the predictor transfer functions $\mathbf{W}_x$ and $\mathbf{W}_r$ and the covariance Σ_ϵ are implicitly parameterized via an algebraic Riccati equation, cf. (Hansson et al., 2025). The signal x is measured in addition to $\bar{r}$. In case $\bar{y}_k^B$ is independent of e_k it holds that this is an ML problem. This is however not the case in general for our problem, and then we have what in the literature is called a PEM problem with an ML-like criterion function, cf. (Ljung, 1978).

We will not solve the problem as it stands, but instead use an equivalent formulation presented in (Da Mata et al., 2024) and based on maximum likelihood estimation for missing data, (Hansson and Wallin, 2012; Wallin and Hansson, 2014). In this way we will not have to consider the algebraic Riccati equation as a constraint, which would have been difficult. However, the predictor formulation we have above will make it possible for us to conclude that we can estimate $\mathbf{G}_A$ consistently.

Consistency of the corresponding estimator for identification of complete networks was established in (Hansson et al., 2025). The predictor used here is the same Kalman filter predictor, but applied to the subnetwork shown in Figure 3, where the signals entering the subnetwork from the surrounding network are treated as additional measured inputs. Consequently, the only difference compared to the setting considered in (Hansson et al., 2025) is that some of the measured inputs originate from the surrounding feedback network.

Theorem 1. Let $\hat{\theta}_N$ be the parameter component of a global minimizer of (5) based on N samples. Suppose that the boundary-information condition holds and that the following conditions are satisfied: (i) the A- and B-polynomials of the ARMAX models are coprime; (ii) the C-polynomials of the ARMAX models have no zeros on the unit circle; (iii) $(x_k, \bar{r}_k)$ is informative enough; and (iv) the A-part of the network is generically network identifiable. Then, by Theorem 7 of (Hansson et al., 2025),

$$\mathbf{G}_A(q;\hat{\theta}_N) \xrightarrow[N\to\infty]{\text{a.s.}} \mathbf{G}_A(q;\theta_0) \tag{6}$$

where a.s. denotes convergence with probability one.

Proof. At the true parameter, the first two stated conditions provide the regularity conditions for the ARMAX predictor. Strict properness of $\mathbf{G}(q)$ prevents the current disturbance in the enlarged A-part from reaching $\bar{y}_k^B$. Diagonality of $\bar{\mathbf{H}}(q)$ and Σ_e separates the current disturbances in the enlarged A-part and the remaining B-part. Finally, the boundary-information condition ensures that every measured boundary signal entering A depends on A only through information reconstructible from past measurements. Consequently, the information used by the predictor contains no current A-part innovation, and

$$\mathbb{E}\left[\epsilon_k x_j^T\right] = 0, \quad j < k, \qquad \mathbb{E}\left[\epsilon_k \bar{r}_j^T\right] = 0, \quad j \le k. \tag{7}$$

The orthogonality relations in (7) are the additional condition needed in the subnetwork setting. The standard prediction-error consistency argument, including the closed-loop case, is given in (Ljung, 1999, Secs. 8.3–8.4 and 13.4–13.5); see also (Forssell and Ljung, 1999). Together with the informativity and identifiability conditions, these relations allow the consistency result in Section VII of (Hansson et al., 2025) to be applied to the enlarged A-part.

Remark 1. For the subvector θ_{G_A} parameterizing $\mathbf{G}_A$, $\hat{\theta}_{G_A,N} \xrightarrow{\text{a.s.}} \theta_{G_A,0}$ as $N \to \infty$. Moreover, without loss of generality, it may be necessary to replace x_k by suitable linearly independent linear combinations of its components in order to satisfy Assumption A2 of (Hansson et al., 2025). See (Hansson et al., 2026) for informativity and generic network identifiability.

The main motivation for considering the PEM formulation above is that it permits predictors that are nonlinear in the open-loop transfer functions in $\mathbf{G}_A$. Such predictors can require fewer measured signals in the A-part than linear predictors while still yielding consistent estimates of the parameters of $\mathbf{G}_A$. It should be mentioned that, in general, we cannot estimate the noise model.

4. NUMERICAL EXAMPLE

4.1 Network and measurement configurations

We consider a seven-module network and estimate $\mathbf{G}^1, \mathbf{G}^2, \mathbf{G}^3$, with $A = \{1,2,3\}$ and $B = \{4,5,6,7\}$. Its interconnection is specified by

$$\begin{aligned} & u^1 = y^6, \quad u^2 = r^1 + y^1 + y^3, \quad u^3 = r^2 + y^2, \\ & u^4 = y^7, \quad u^5 = y^4, \qquad\qquad u^6 = r^3 + y^5 + y^7, \\ & u^7 = y^3. \end{aligned} \tag{8}$$

For this example, $\mathbf{\Omega}_y = \mathbf{H}_u = 0$, $\mathbf{\Upsilon} = \Upsilon$, and $\mathbf{\Omega}_u = \Omega$. The interconnection matrices in (2) for (8) are

$$\Upsilon = \begin{bmatrix} 0&0&0&0&0&1&0 \\ 1&0&1&0&0&0&0 \\ 0&1&0&0&0&0&0 \\ 0&0&0&0&0&0&1 \\ 0&0&0&1&0&0&0 \\ 0&0&0&0&1&0&1 \\ 0&0&1&0&0&0&0 \end{bmatrix}, \quad \Omega = \begin{bmatrix} 0&0&0 \\ 1&0&0 \\ 0&1&0 \\ 0&0&0 \\ 0&0&0 \\ 0&0&1 \\ 0&0&0 \end{bmatrix}. \tag{9}$$

In the notation of the preceding section, the incoming signal is $\bar{y}^B = \Upsilon_{AB} y^B = (y^6, 0, 0)$. The external input r^3 does not enter A directly. Omitting these unused and zero components, the measured input signal for estimation is (r^1, r^2, y^6), and

$$\begin{aligned} \Upsilon_A &= \begin{bmatrix} 0&0&0 \\ 1&0&1 \\ 0&1&0 \end{bmatrix}, \\ \Omega_A r_k + \Upsilon_{AB} y_k^B &= \begin{bmatrix} 0&0&1 \\ 1&0&0 \\ 0&1&0 \end{bmatrix} \begin{bmatrix} r_k^1 \\ r_k^2 \\ y_k^6 \end{bmatrix}. \end{aligned} \tag{10}$$

Table 1. Data-generating coefficients; $\lambda^i = 0.01i$.

i	a_1^i	a_2^i	b_1^i	b_2^i	c_1^i	c_2^i
1	1	0.25	0.3	0.16	0.3	-0.01
2	-0.8	0.15	0.8	-0.3	-0.8	0.2
3	0.45	-0.13	-0.4	-0.25	-0.02	-0.8
4	-0.45	-0.1	-2	0.4	-0.15	-0.07
5	0.1	-0.4	2.2	2	-0.6	-0.05
6	-0.2	-0.15	0.15	0.05	1	0.15
7	0.5	0.05	1	0.2	0.15	-0.7

The measured output vector x_k is y_k^3, (y_k^1, y_k^3), or (y_k^1, y_k^2, y_k^3) in the three configurations considered; the same measured inputs are available in every case. The signal from A to B that can return to A is y_k^3, which is included in every choice of x_k. Hence the selected A-part satisfies the simple structural sufficient condition, and no enlargement is needed in this example.

For the diagram, write $v^i = \mathbf{H}^i e^i$ for each module disturbance.

All modules have order two, with transfer functions

$$\mathbf{G}^i(q) = \frac{b_1^i q + b_2^i}{q^2 + a_1^i q + a_2^i}, \qquad \mathbf{H}^i(q) = \frac{q^2 + c_1^i q + c_2^i}{q^2 + a_1^i q + a_2^i}. \tag{11}$$

The plant coefficients are $a_1^i, a_2^i, b_1^i, b_2^i$; the coefficients c_1^i, c_2^i describe the noise numerator, and λ^i is the variance of e_k^i.

4.2 Algebraic recovery and identifiability

For the coefficients in Table 1, the poles of the full network and of subnetwork A, with (r^1, r^2, y^6) treated as external inputs, have absolute values less than 0.91850 and 0.64341, respectively. All seven plant modules are stable and strictly proper. Eliminating the unmeasured outputs y^1 and y^2 gives

$$\begin{aligned}\Delta y^3 = {} & \mathbf{G}^3\mathbf{G}^2\mathbf{G}^1 y^6 + \mathbf{G}^3\mathbf{G}^2 r^1 + \mathbf{G}^3 r^2 \\ & + \mathbf{G}^3\mathbf{G}^2\mathbf{H}^1 e^1 + \mathbf{G}^3\mathbf{H}^2 e^2 + \mathbf{H}^3 e^3,\end{aligned} \tag{12}$$

where $\Delta = 1 - \mathbf{G}^3\mathbf{G}^2$ and transfer operators act on the associated signals. In the input ordering (y^6, r^1, r^2), the transfer matrix from measured inputs to y^3 has entries

$$T_1 = \frac{\mathbf{G}^3\mathbf{G}^2\mathbf{G}^1}{\Delta}, \qquad T_2 = \frac{\mathbf{G}^3\mathbf{G}^2}{\Delta}, \qquad T_3 = \frac{\mathbf{G}^3}{\Delta}. \tag{13}$$

The target modules are recovered by

$$\mathbf{G}^1 = T_1/T_2, \qquad \mathbf{G}^2 = T_2/T_3, \qquad \mathbf{G}^3 = T_3/(1+T_2). \tag{14}$$

Equations (14) establish generic uniqueness, or generic network identifiability from (T_1, T_2, T_3) whenever the denominators are not identically zero. For coprime plant polynomials with monic denominators of the prescribed orders, the transfer functions in (14) also determine the plant coefficients. Thus the three target plants can be recovered from the transfer functions between the measured inputs and y^3, even when y^1 and y^2 are unmeasured.

Direct linear PEM for all three open-loop transfer functions requires measuring (y^1, y^2, y^3) together with the measured signals (y^6, r^1, r^2). This gives the equations

$$\begin{aligned} y^1 &= \mathbf{G}^1 y^6 + \mathbf{H}^1 e^1, \\ y^2 &= \mathbf{G}^2(r^1 + y^1 + y^3) + \mathbf{H}^2 e^2, \\ y^3 &= \mathbf{G}^3(r^2 + y^2) + \mathbf{H}^3 e^3. \end{aligned} \tag{15}$$

In (15), every open-loop transfer function has a fully measured input. With fewer internal measurements, direct PEM will be nonlinear in the open-loop transfer functions.

4.3 Estimation, validation, and baseline

The estimation data contain $N = 500$ samples with independent Gaussian disturbances of variances $\lambda^i = 0.01i$. The external signals are random binary sequences taking values -1 and 1, generated using MATLAB's `idinput` with the `rbs` option. For parameter statistics, 100 disturbance realizations use the same external-input data. Thus these statistics describe variability conditional on that excitation data.

We do not impose the algebraic Riccati equation as an optimization constraint. Instead, we use the equivalent missing-data formulation of (Da Mata et al., 2024), based on (Hansson and Wallin, 2012; Wallin and Hansson, 2014). The implementation uses three initialization stages: autoregressive estimation with exogenous inputs (ARX) and a common disturbance variance, ARMAX estimation with that common variance, and ARMAX estimation with separate variances. It uses a trust-region method from SciPy's `optimize` package, with derivatives computed using JAX (Bradbury et al., 2018), double precision, default SciPy stopping tolerances, and at most 1000 iterations per stage. The second and third stages are initialized with the estimate from the preceding stage. Positive noise variances are parameterized as $\lambda^i = 10^{-2\rho_i}$ with unconstrained ρ_i.

The baseline is direct PEM, implemented through separate ARMAX estimation in MATLAB R2024a using the System Identification Toolbox (Ljung, 1995). The three equations in (15) are fitted separately using `armax` with orders $[n_a, n_b, n_c, n_k] = [2, 2, 2, 1]$ and `SearchOptions.Tolerance`$= 10^{-5}$. Here n_a, n_c are polynomial orders, n_b counts input coefficients, and n_k is the input delay. With sparse measurement, not all required input/output pairs are available; this baseline is therefore used only when all three outputs are measured.

For validation, a separate 500-sample data record supplies r^1, r^2, y^6 to the estimated subnetwork. Plant-only simulations $\hat{y}^i$, without new internal disturbances, are compared with noisy reference outputs $y^{i,\mathrm{ref}}$ using

$$F_i = 100\left(1 - \frac{\|\hat{y}^i - y^{i,\mathrm{ref}}\|_2}{\|y^{i,\mathrm{ref}} - \mathbf{1}\bar{y}^{i,\mathrm{ref}}\|_2}\right), \tag{16}$$

where $\bar{y}^{i,\mathrm{ref}}$ is the sample mean and $\mathbf{1}$ is the 500-dimensional vector of ones. These individual simulation fits use measured inputs, not one-step predictions or free full-network simulation. Withheld outputs are used only for scoring. Both methods use the same 500 validation samples without burn-in removal.

For the Monte Carlo statistics, define the target coefficient vector as $\alpha_A = (a_1^1, a_2^1, b_1^1, b_2^1, \ldots, a_1^3, a_2^3, b_1^3, b_2^3) \in \mathbb{R}^{12}$. Let $\hat{\alpha}^{(\ell)}$ denote its estimate in run ℓ. Let $\alpha_{A,0}$ be its true value. For $R = 100$ runs, define the sample mean $\bar{\alpha}$, empirical

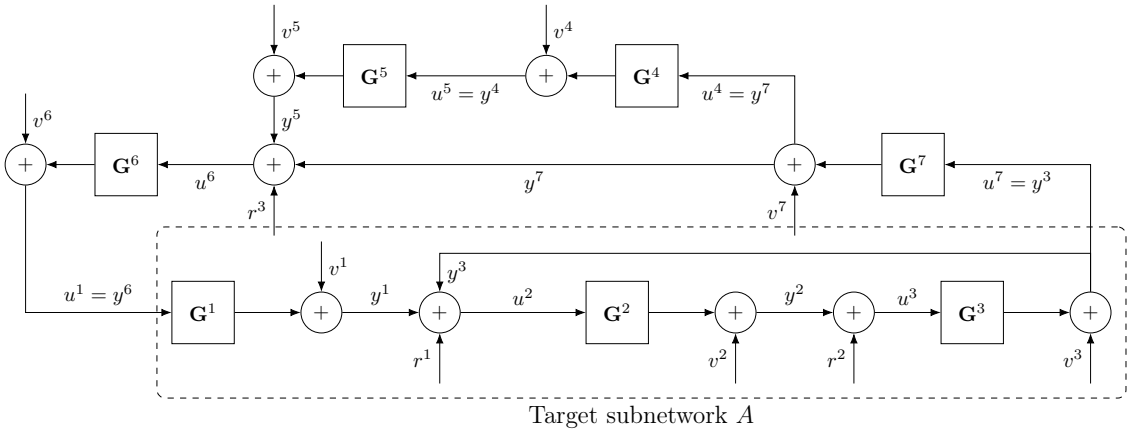


Fig. 4. Example network, with target modules $A = \{1, 2, 3\}$ enclosed by the dashed line and remaining modules in B. The signal y^6 is measured in every experiment. Internal measurements are y^3, (y^1, y^3), or (y^1, y^2, y^3).

covariance $\widehat{C}$, squared bias B^2, and mean-square error (MSE) by

$$\bar{\alpha} = R^{-1} \sum_{\ell=1}^{R} \hat{\alpha}^{(\ell)},$$

$$\widehat{C} = R^{-1} \sum_{\ell=1}^{R} (\hat{\alpha}^{(\ell)} - \bar{\alpha})(\hat{\alpha}^{(\ell)} - \bar{\alpha})^T,$$

$$B^2 = \|\bar{\alpha} - \alpha_{A,0}\|_2^2, \qquad \text{MSE} = \operatorname{tr} \widehat{C} + B^2. \tag{17}$$

The divisor R makes (17) exact. All saved estimates are included, without stability-based exclusion. Code and results are public. [2]

4.4 Results and discussion

For the network in Figure 4, the proposed PEM and direct PEM give similar results with the same measured signals. Its main advantage is that all three open-loop transfer functions can be recovered using fewer internal measurements, with only a small decrease in the fit values: measuring only y^3 changes each fit by less than one percentage point relative to complete internal measurement.

A larger parameter bias is observed when fewer signals are measured. The squared bias increases from 0.1974 with all three outputs to 1.3776 with y^3 alone, while the parameter MSE increases from 1.0475 to 1.7539. The objective in (5) is nonconvex, so convergence to local minima can contribute to the observed parameter bias.

5. CONCLUSIONS

This paper has shown that a class of subnetwork identification problems can be reformulated as complete-network identification problems through augmentation of the measured external signals. By augmenting the measured exogenous signals with measured boundary signals entering the subnetwork from the surrounding network, the subnetwork identification problem can be cast in a form closely related to the identification of a complete network.

Consistency requires the A-generated component of each measured boundary signal entering the A-part to be reconstructible from the measured history. Measuring every signal from A entering B is a simple sufficient condition, but it is not necessary. The A-part may be enlarged to satisfy this structural condition, although reconstructibility can permit a smaller subnetwork. The same predictor construction as in (Hansson et al., 2025) can then be applied to the enlarged A-part.

An important advantage of the approach is that it allows the use of predictors that are nonlinear in the open-loop transfer functions. This makes it possible to obtain consistent estimates using fewer measured signals than what is typically required by PEM methods based on predictors that are linear in the open-loop transfer functions.

Future work will focus on numerical aspects of the optimization problem and further investigations of measurement requirements for identification of subnetworks.

[2] `https://github.com/Jvictormata/mle_sub_nets`

6. APPENDIX

Consider (1)–(2), and assume that $m_e \leq n$. For $p \in \{n, m_r\}$, define

$$J_p = \mathbb{1}_n \otimes I_p \in \mathbb{R}^{np \times p}, \qquad S_p = I_n \otimes \mathbb{1}_p^T \in \mathbb{R}^{n \times np}, \tag{18}$$

where $\mathbb{1}_p$ is the p-dimensional vector of ones. For a transfer matrix $\mathbf{A}(q) \in \mathbb{R}^{n \times p}(q)$, let $\operatorname{vec}_{\mathrm{r}}(\mathbf{A})$ denote the vector obtained by stacking the entries of $\mathbf{A}$ row by row, and define

$$D_{\mathbf{A}}(q) = \operatorname{diag}(\operatorname{vec}_{\mathrm{r}}(\mathbf{A}(q))) \in \mathbb{R}^{np \times np}(q). \tag{19}$$

The definitions in (18)– (19) give

$$\mathbf{A}(q) z_k = S_p D_{\mathbf{A}}(q) J_p z_k, \qquad z_k \in \mathbb{R}^p. \tag{20}$$

Let $d = n^2 + n m_r$ and define

$$\begin{aligned} e_k^p &= \begin{bmatrix} e_k \\ 0_{n-m_e} \end{bmatrix} \in \mathbb{R}^n, \\ \mathbf{H}_y^p &= \begin{bmatrix} \mathbf{H}_y & 0_{n \times (n-m_e)} \end{bmatrix}, \\ \mathbf{H}_u^p &= \begin{bmatrix} \mathbf{H}_u & 0_{n \times (n-m_e)} \end{bmatrix}. \end{aligned} \tag{21}$$

Then $\mathbf{H}_y^p e_k^p = \mathbf{H}_y e_k$ and $\mathbf{H}_u^p e_k^p = \mathbf{H}_u e_k$. Define

Table 2. Individual simulation fits and 100-run parameter statistics; measured inputs: y^6, r^1, r^2.

Internal outputs	Method	F_1	F_2	F_3	$\operatorname{tr}\widehat{C}$	B^2	MSE
y^3	Proposed PEM	56.58	75.89	60.36	0.3764	1.3776	1.7539
y^1, y^3	Proposed PEM	57.10	75.84	60.21	0.2442	1.2648	1.5090
y^1, y^2, y^3	Proposed PEM	57.13	76.86	60.43	0.8501	0.1974	1.0475
y^1, y^2, y^3	Direct PEM (linear in $\mathbf{G}$)	57.13	76.86	60.43	0.8717	0.1400	1.0117

$$\tilde{\mathbf{G}} = \operatorname{blkdiag}\big(D_{\mathbf{G}}, D_{\mathbf{\Omega}_y}, D_{\mathbf{\Upsilon}}, D_{\mathbf{\Omega}_u}\big) \in \mathbb{R}^{2d\times 2d}(q), \quad (22)$$

$$\tilde{\mathbf{H}} = \operatorname{blkdiag}\Big(D_{\mathbf{H}_y^p}, 0_{nm_r\times nm_r}, D_{\mathbf{H}_u^p}, 0_{nm_r\times nm_r}\Big) \in \mathbb{R}^{2d\times 2d}(q), \quad (23)$$

$$\tilde{e}_k = \begin{bmatrix} J_n e_k^p \\ 0_{nm_r} \\ J_n e_k^p \\ 0_{nm_r} \end{bmatrix} \in \mathbb{R}^{2d}. \quad (24)$$

Both $\tilde{\mathbf{G}}$ and $\tilde{\mathbf{H}}$ are diagonal transfer matrices.

Define

$$P = [S_n \;\; S_{m_r}] \in \mathbb{R}^{n\times d}, \qquad K = J_n P \in \mathbb{R}^{n^2\times d}. \quad (25)$$

Using block-row dimensions n^2, nm_r, n^2, nm_r and two block-column dimensions d, d, let

$$\tilde{\Upsilon} = \begin{bmatrix} 0_{n^2\times d} & K \\ 0_{nm_r\times d} & 0_{nm_r\times d} \\ K & 0_{n^2\times d} \\ 0_{nm_r\times d} & 0_{nm_r\times d} \end{bmatrix} \in \mathbb{R}^{2d\times 2d}, \quad (26)$$

$$\tilde{\Omega} = \begin{bmatrix} 0_{n^2\times m_r} \\ J_{m_r} \\ 0_{n^2\times m_r} \\ J_{m_r} \end{bmatrix} \in \mathbb{R}^{2d\times m_r}. \quad (27)$$

The expanded network is

$$\tilde{y}_k = \tilde{\mathbf{G}}(q)\tilde{u}_k + \tilde{\mathbf{H}}(q)\tilde{e}_k, \quad (28)$$

$$\tilde{u}_k = \tilde{\Upsilon}\tilde{y}_k + \tilde{\Omega} r_k. \quad (29)$$

Finally, with

$$\Psi = \operatorname{blkdiag}(P, P) \in \mathbb{R}^{2n\times 2d}, \quad (30)$$

equation (20) gives

$$\begin{bmatrix} y_k \\ u_k \end{bmatrix} = \Psi \tilde{y}_k. \quad (31)$$

Thus, (1)–(2) and (28)–(31) generate the same signals y_k and u_k. The expanded description has diagonal plant and noise transfer matrices and is a Box–Jenkins description. If a component of $\tilde{y}_k$ is identically zero, that component and the corresponding rows and columns can be removed.

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